PhysMoE: Fast Spatial Grounding with Single-Token Coordinate Representation in Modality-Decoupled VLMs

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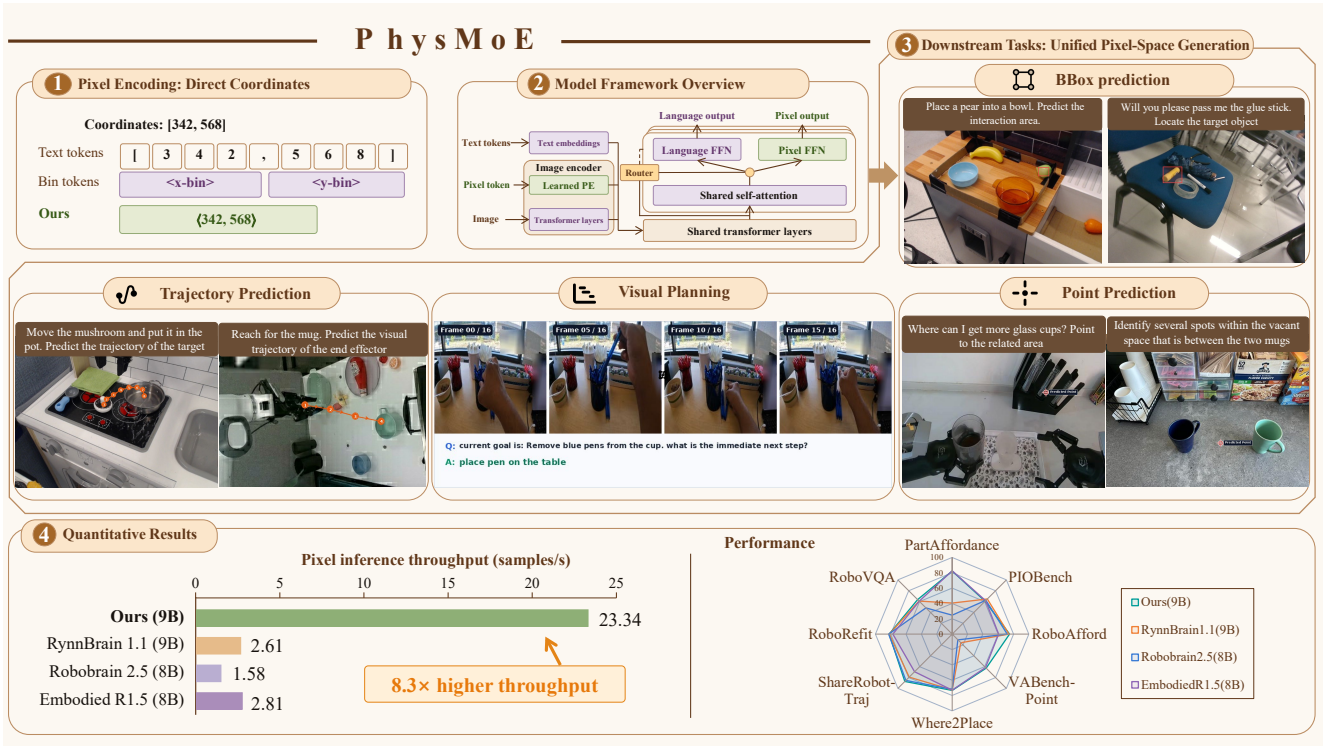


Figure 1. **Overview of the PhysMoE framework**. (a) Comparison between conventional coupled architectures and our decoupled approach. (b) The proposed PhysMoE framework featuring continuous pixel position encoding, Semi-MoE token routing with decoupled linguistic (FFN_{lang}) and spatial (FFN_{pix}) experts, and Mask-SDF supervision.

Abstract

*Embodied Vision-Language Models (VLMs) are backbones for robotic planning and manipulation. However, for coordinate prediction in grounding tasks, existing architectures suffer from two distinct bottlenecks: discrete tokenization disrupts spatial continuity with high decoding latency, while standard point regression fails on multi-contact affordances and irregularly shaped targets. To resolve these challenges, we propose **PhysMoE**, introducing three core contributions: (1) sharing lower layers for multimodal context while routing upper layers to specialized spatial (pixel) and linguistic (text) experts by physical modality, eliminating gradient interference; (2) encoding each coordinate as a single token by reusing the visual encoder's learned positional embeddings, injecting 2D spatial priors to natively align coordinates with visual features and ease training; and (3) differentiable Signed Distance Field (SDF) supervision on pointing, accommodating affordances and irregular geometries without costly mask decoders. Compared to the*

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base model, PhysMoE achieves a 15.7% average gain and an $8.31\times$ throughput speedup across 9 coordinate-based grounding benchmarks (trajectory, affordance, and localization), while preserving multimodal understanding and reasoning across 11 text-based embodied benchmarks.

1. Introduction

Embodied Vision-Language Models (VLMs) have emerged as powerful backbones for robotic planning and control [6, 7, 15, 65]. An effective embodied agent must not only understand language instructions and reason over visual scenes, but also ground decisions into fine-grained 2D coordinates—such as identifying grasp contact points, object placement regions, and dynamic manipulation trajectories [30, 59].

Despite rapid advances, existing autoregressive VLMs struggle to balance continuous spatial manipulation with high-level multimodal reasoning due to two fundamental bottlenecks:

1. Fragmented Spatial Tokenization and Modality Interference in Dense Decoders. Mainstream architectures represent coordinates either as character-level text strings [6, 8] or discrete spatial bin tokens [38]. Under standard VLM sub-word tokenizers, a single coordinate pair such as `"[342, 518]"` is partitioned into a sequence of isolated character and delimiter tokens (*e.g.*, `[`, `3`, `4`, `2`, `,`, `5`, `1`, `8`, `]`). This representation breaks the holistic spatial integrity of pixel positions and imposes an unnecessary representational burden on semantic understanding. Moreover, generating coordinates autoregressively requires 8 to 10 sequential decoding passes per point, severely inflating inference latency and introducing quantization errors. When these disparate tasks are processed through standard dense decoders where all tokens share identical Feed-Forward Networks (FFNs), high-magnitude spatial regression gradients interfere with pre-trained linguistic weights, creating modality conflicts between categorical text generation and geometric modeling.

2. Multi-Peak Affordance Ambiguity. Physical affordances rarely correspond to an isolated, unique point. For tasks such as grasping a utensil handle or placing a bowl onto a table, multiple contact locations across the object part are equally functional, forming a distinct *multi-peak distribution*. Conventional point grounding enforces single-point regression (e.g., ℓ_1 or ℓ_2 distance to an arbitrary center). This formulation is fundamentally mismatched with multi-peak targets: it penalizes the model for predicting any valid contact mode away from the center and often forces the network to regress to a blurred average of multiple modes that lands in invalid physical space. Conversely, full mask prediction models like Qwen3-VL-Seg [57] introduce substantial compute and memory overhead.

To resolve these challenges, we propose **PhysMoE**, an embodied vision-language model featuring three synergistic innovations: Firstly, we represent a 2D coordinate as a unified token $\langle|x, y|\rangle$ and derive their representations by reusing the pre-trained learned positional embeddings of the visual encoder. This preserves holistic 2D spatial integrity and simplifies model training. Secondly, we keep the lower decoder layers fully shared, and in the top K decoder layers we preserve shared attention across all tokens for unified scene context while routing tokens between a linguistic expert (FFN_{lang}) and a specialized spatial expert (FFN_{pix}) paired with decoupled output heads (a continuous coordinate regression head and a discrete LM head for language), resolving modality interference. Thirdly, we formulate a differentiable Euclidean Signed Distance Field (ESDF) loss to accommodate multi-peak affordances without auxiliary mask decoders. By sampling continuous coordinates directly from target ESDF maps built from binary masks, our asymmetric loss rewards points falling anywhere inside valid affordance regions while providing steep restorative gradients for exterior predictions.

Our main contributions are summarized as follows:

1. We encode each 2D coordinate as a single token by reusing the visual encoder's learned positional embeddings, aligning coordinates with visual features to ease training.
2. We propose a **PhysMoE** architecture that shares lower layers while routing the top K layers to modality-specific linguistic and spatial experts, eliminating gradient interference.
3. We introduce differentiable ESDF supervision on pointing, accommodating multi-contact affordances and irregular targets without mask decoders.
4. Across 20 benchmarks, PhysMoE achieves a 15.7% average gain and an $8.31\times$ speedup on 9 coordinate-based grounding tasks while preserving reasoning across 11 text-based embodied benchmarks.

2. Related Work

Embodied Vision-Language Models. Multimodal foundation models [2, 6, 39, 48] have inspired a surge of embodied VLMs for robotic planning and control [7, 15, 65]. Backbones such as RoboBrain [46], RynnBrain [30], HY-Embodied-VLM [23], and Embodied-R1.5 [61] advance spatial scene understanding and physical reasoning. However, these systems typically handle spatial coordinates either through discrete text generation or separate downstream policies. In contrast, PhysMoE unifies continuous physical manipulation and conversational reasoning within an autoregressive architecture via an expert-decoupled design.

Coordinate Encodings in Multimodal Backbones. Spatial coordinates are conventionally tokenized as

character-level text strings (*e.g.*, partitioned into individual tokens `[`, `3`, `4`, ...) [6, 8, 52] or discretized spatial bins [10, 38]. While compatible with standard vocabularies, discrete encodings suffer from artificial quantization errors, sequence bloat, and language degradation during fine-tuning. Recently, Molmo [13] demonstrated point pointing via direct coordinate prediction. Expanding on this direction, our ablation studies reveal that *reusing the pre-trained learned 2D positional embeddings* of the visual encoder via bilinear interpolation yields superior spatial accuracy and faster optimization than training coordinate embeddings from scratch or using fixed sinusoidal Fourier encodings.

Modality Decoupling and Multi-Peak Affordance Learning. Mixture of Experts (MoE) [12, 18, 25, 44] is widely used to scale capacity. In multimodal contexts [31, 32], experts typically route across general data-driven clusters. In contrast, our **PhysMoE** enforces a structural division of labor: self-attention remains shared across all tokens to preserve unified context, while top-layer FFNs disjunctively isolate linguistic generation (FFN_{lang}) from continuous spatial regression (FFN_{pix}). Furthermore, physical affordance learning [14, 20, 35] is inherently a *multi-peak problem*, where an actionable surface contains multiple valid contact points. Standard single-point regression penalizes valid alternate modes or averages them into invalid regions. Inspired by implicit signed distance fields in 3D geometry [37] and robotics [36, 40], we introduce 2D Truncated Euclidean Signed Distance Fields (ESDF) as a continuous, differentiable loss. This resolves the multi-peak ambiguity by softly supervising regional affordances without introducing heavy mask decoders (e.g., SAM [26]).

3. Methodology

In this section, we present the architecture and training methodology of PhysMoE. We first detail the Semi-MoE decoupled architecture with dynamic routing (Sec. 3.1), describe the continuous pixel position representation and bilinear positional encoding (Sec. 3.2), introduce the differentiable Mask-SDF map supervision for multi-peak targets (Sec. 3.3), and outline the two-stage training curriculum (Sec. 3.4).

3.1. Semi-MoE Decoupled Architecture

Standard dense transformers process all tokens through identical Feed-Forward Networks (FFNs), causing acute gradient interference between discrete categorical text classification and continuous coordinate regression. To resolve this conflict while retaining global visual-linguistic context, we introduce the **Semi-MoE** architecture (Fig. 2).

Layer Partitioning and Shared Token Mixer. Let the Transformer backbone comprise L sequential layers ($l = 0, \dots, L-1$). We partition the decoder into two functional stages:

- *Shared Context Layers ($l = 0, \dots, L-K-1$)*: The initial $L-K$ layers operate as standard dense Transformer blocks where visual, textual, and spatial tokens attend to one another freely, building rich multimodal contextual representations.
- *Semi-MoE Routed Layers ($l = L-K, \dots, L-1$)*: Across the top K layers (e.g., $K = 8$), we maintain the shared self-attention mixer across all tokens to preserve uninterrupted global receptive fields. However, the subsequent feed-forward computation is decoupled into specialized experts.

Dynamic Token Router. At layer $L-K$, a lightweight router MLP $\mathcal{R}$ inspects the normalized hidden state $\mathbf{h}_t \in \mathbb{R}^{d_{\text{text}}}$ and predicts binary routing logits:

$$\mathbf{r}_t = \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\mathbf{h}_t + \mathbf{b}_1) + \mathbf{b}_2 \in \mathbb{R}^2, \tag{1}$$

where $r_{t,0}$ and $r_{t,1}$ denote the routing propensities for linguistic versus spatial coordinate generation, respectively. During training, routing is teacher-forced using the ground-truth modality $c_t^* \in \{0, 1\}$:

$$\mathcal{L}_{\text{router}} = \text{CE}_{\text{eos}}\big(\mathbf{r}_t, c_t^*; w_{\text{eos}}\big), \tag{2}$$

weighted by an End-of-Sequence factor w_{eos}. During inference, hard discrete gating is executed: $c_t = \arg\max(\mathbf{r}_t)$.

Modality-Specialized Experts. In the top K routed layers, after shared self-attention, each token is processed by a modality-specific feed-forward network according to c_t: linguistic tokens ($c_t = 0$) pass through FFN_{lang}, whereas spatial tokens ($c_t = 1$) are routed to FFN_{pix}. This decoupling isolates continuous spatial regression gradients from textual semantic representations.

Decoupled Dual Prediction Heads. At the final layer L, hidden states are mapped to predictions via two task-specific heads:

- *LM Head*: A linear projection $\mathbf{W}_{\text{lm}} \in \mathbb{R}^{d_{\text{text}} \times V}$ predicting logits over vocabulary V, supervised via cross-entropy:

$$\mathcal{L}_{\text{LM}} = \text{CE}_{\text{eos}}\big(\mathbf{W}_{\text{lm}}\mathbf{h}_t, y_{t+1}^*; w_{\text{eos}}\big). \tag{3}$$

- *Pixel Head* : An MLP that directly regresses continuous normalized 2D coordinates:

$$\hat{\mathbf{p}}_t = \sigma\Big(\mathbf{W}_{\text{pix},2}\,\text{GELU}(\mathbf{W}_{\text{pix},1}\mathbf{h}_t + \mathbf{b}_{\text{pix},1}) + \mathbf{b}_{\text{pix},2}\Big) \in [0, 1]^2, \tag{4}$$

where $\sigma(\cdot)$ is the Sigmoid activation enforcing normalized coordinate bounds $[0, 1]$.

3.2. Continuous Pixel Position Representation and Encoding

Representation Syntax. Within conversational dialogues and instructions, 2D coordinates are formatted as $\langle|x, y|\rangle$,

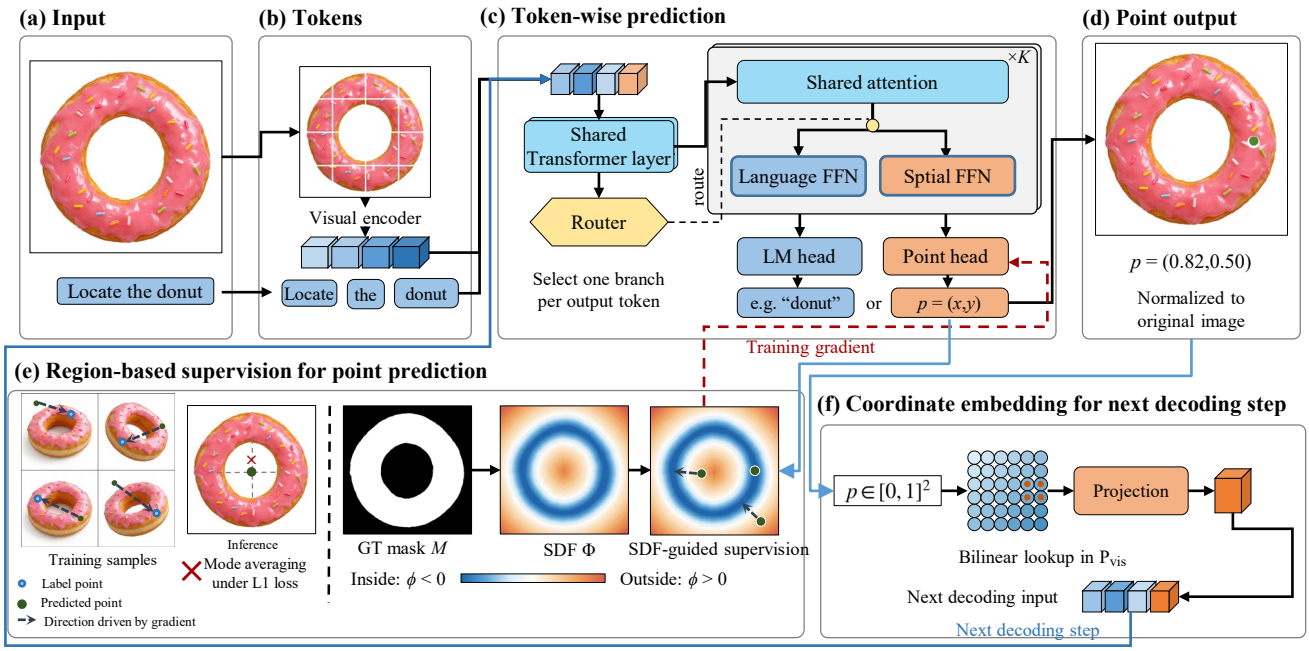


Figure 2. **PhysMoE Architecture, Region-Based Supervision, and Continuous Coordinate Embedding**. (a) Input RGB image and prompt. (b) Sequence tokenization. (c) Semi-MoE layers with shared self-attention and dynamically routed language/spatial experts (FFN_{lang}, FFN_{pix}). (d) Decoupled prediction heads for language tokens and 2D coordinates. (e) Region-based Mask-SDF supervision for continuous point prediction, penalizing exterior predictions while accommodating multi-peak affordance regions without mode averaging. (f) Continuous coordinate embedding via bilinear interpolation over visual positional embeddings $\mathcal{P}_{\text{vis}}$ and linear adapter projection for subsequent decoding steps.

where $x, y \in [0, 1]$ are floating-point values normalized by image width and height. Our custom tokenizer maps each coordinate to a single unified token, which preserves prompt conciseness in inputs and bypasses the multi-token autoregressive decoding traditionally required for coordinate serialization. Furthermore, treating coordinates as unified geometric units allows the tokenizer to flexibly support arbitrary pixel delimiter tokens (*e.g.*, brackets, parentheses, or task-specific tags) without altering model weights.

Coordinate Encoding via Visual Positional Embeddings. To embed continuous coordinates $(x, y) \in [0, 1]^2$ into dense vectors that share the visual encoder's spatial frame, we reuse the pre-trained learned 2D positional embedding grid $\mathcal{P}_{\text{vis}} \in \mathbb{R}^{G \times G \times d_{\text{vis}}}$ of the Vision Transformer (e.g., $G = 48$). Rather than quantizing coordinates into discrete grid bins, we query $\mathcal{P}_{\text{vis}}$ via bilinear interpolation and project the resulting feature into the language hidden dimension $\mathbb{R}^{d_{\text{text}}}$ with a linear layer (Fig. 2f):

$$\mathbf{e}_{\text{pix}}(x, y) = \mathbf{W}_{\text{embed}}\,\text{BilinearInterpolate}\big(\mathcal{P}_{\text{vis}},\,(x, y)\big) + \mathbf{b}_{\text{embed}}. \tag{5}$$

As demonstrated in our ablation studies (Sec. 4.3), reusing the visual encoder's learned positional embeddings provides an essential geometric inductive bias that natively aligns coordinates with visual token representations and eases training.

3.3. Differentiable ESDF Supervision for Point Prediction

Resolving Multi-Peak Ambiguity in Affordance Grounding. In real-world robotic manipulation, actionable affordances and placement regions span continuous 2D zones $M \subset [0, 1]^2$, forming an inherently *multi-peak distribution* with multiple valid contact locations. Conventional point grounding enforces single-point regression (e.g., ℓ_1 or Smooth L1 distance to an arbitrary mask centroid). This penalizes the model for predicting any valid contact mode away from the center and often forces it to output an arithmetic centroid that lands outside valid physical space. To resolve this without costly mask decoders, we propose **ESDF supervision** (Fig. 2e).

Euclidean Signed Distance Field Formulation. As illustrated in Fig. 2e, for a target affordance region $M \subset [0, 1]^2$ enclosed by boundary contour ∂M, the signed distance $\phi_M(p)$ of any 2D point $p \in [0, 1]^2$ measures its Euclidean distance to the nearest boundary:

$$\phi_M(p) = \begin{cases} -\min_{q \in \partial M} \|p - q\|_2, & \text{if } p \in M \text{ (interior)}, \\ +\min_{q \in \partial M} \|p - q\|_2, & \text{if } p \notin M \text{ (exterior)}, \end{cases} \tag{6}$$

where $q \in \partial M$ represents a point on the mask boundary ∂M. Thus, $\phi_M(p) < 0$ inside the mask, $\phi_M(p) = 0$ on the boundary, and $\phi_M(p) > 0$ outside.

Differentiable Bilinear Grid Sampling. Given the predicted continuous coordinate $\hat{\mathbf{p}} = (\hat{x}, \hat{y}) \in [0, 1]^2$ emitted by the continuous coordinate head, we sample the scalar signed distance $\phi(\hat{\mathbf{p}})$ directly from the target ESDF map $\Phi \in \mathbb{R}^{S \times S}$ via bilinear grid sampling:

$$\phi(\hat{\mathbf{p}}) = \text{grid_sample}\big(\Phi,\, 2\hat{\mathbf{p}} - 1;\, \text{align_corners} = \text{True}\big). \tag{7}$$

Because `grid_sample` is differentiable with respect to sampling coordinates, gradients $\frac{\partial \phi}{\partial \hat{\mathbf{p}}}$ flow directly into $\hat{\mathbf{p}}$, enabling end-to-end backpropagation without discrete approximation.

Asymmetric Continuous Loss Objective. We formulate an asymmetric loss $\mathcal{L}_{\text{sdf}}$ over the signed distance ϕ:

$$\mathcal{L}_{\text{sdf}}(\phi) = \begin{cases} 0.2 \cdot \exp(25.0 \cdot \phi), & \text{if } \phi \leq 0 \text{ (inside)}, \\ 0.2 + 5.0 \cdot \phi + 8.0 \cdot \phi^2, & \text{if } \phi > 0 \text{ (outside)}. \end{cases} \tag{8}$$

This formulation ensures C^1 smoothness at the boundary $\phi = 0$. For exterior points ($\phi > 0$), the linear-quadratic penalty exerts an increasingly strong inward pull as distance grows. Crucially, assigning a small, exponentially decaying loss to interior points ($\phi \leq 0$) gently pushes predictions near the boundary further inward for a safer margin and more robust localization, while rapidly vanishing with depth to accommodate arbitrary interior contact peaks without center-seeking bias.

Joint Training Objective. The total multi-task objective combines language modeling, routing supervision, and coordinate loss:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \mathcal{L}_{\text{router}} + \lambda_{\text{coord}} \mathcal{L}_{\text{coord}} + \lambda_{\text{esdf}} \mathcal{L}_{\text{sdf}}, \tag{9}$$

where $\mathcal{L}_{\text{coord}}$ denotes Smooth L1 loss applied to fixed coordinate targets (e.g., trajectories), $\mathcal{L}_{\text{sdf}}$ supervises regional affordance masks, and $\lambda_{\text{coord}}, \lambda_{\text{esdf}}$ balance the terms.

3.4. Two-Stage Training Curriculum

Model Initialization. We instantiate PhysMoE at two scales: our 4B model is initialized from Qwen3.5-4B, and our 9B model is initialized from RynnBrain 1.1-9B [30] (which itself is initialized from Qwen3.5-9B). Across the top $K = 8$ semi-MoE layers, the spatial expert FFN_{pix} is initialized by copying weights from the corresponding pre-trained language FFN_{text}, whereas the remaining newly introduced spatial modules—including the router $\mathcal{R}$, continuous coordinate regression head, and coordinate projection MLP ($\text{MLP}_{\text{pix_embed}}$)—are randomly initialized.

Stage 1: Expert Cold-Start and Routing Warmup. To prevent randomly initialized spatial heads and routers from destabilizing the pre-trained backbone, we first freeze the vision encoder, merger, and token embeddings. We apply LoRA to the bottom Transformer layers (layers 0–3) while fully training the spatial modules (the continuous coordinate head, $\text{MLP}_{\text{pix_embed}}$, $\mathcal{R}$, and FFN_{pix}) to stabilize routing boundaries and align continuous coordinate embeddings.

Stage 2: Full Multi-Layer LoRA and Spatial Specialization. In Stage 2, we merge the Stage 1 checkpoint and expand LoRA adaptation across all language layers while continuing full fine-tuning of FFN_{pix}, $\mathcal{R}$, and the coordinate regression head over our full multimodal mixture, yielding our final PhysMoE model.

4. Experiments

In this section, we present an extensive empirical evaluation of PhysMoE. Following the evaluation protocol established in the project benchmark suite, we assess our models across 20 diverse benchmarks comprising 9 coordinate-based spatial grounding tasks and 11 text-based embodied understanding and reasoning tasks.

4.1. Experimental Setup

Training Data Preparation. As summarized in Tab. 1, we curate a massive and balanced multi-task training mixture of **12 million (12M)** samples across two complementary domains: general multimodal cognition and embodied physical grounding. For spatial grounding datasets that initially provide only sparse point or bounding-box (bbx) annotations, we prompt SAM3 [26, 41] with the ground-truth annotations to automatically extract high-quality binary object and affordance masks. These binary masks are subsequently transformed into continuous ESDF maps to provide dense boundary-aware supervision under our Mask-SDF objective (Sec. 3.3).

Evaluation Benchmarks and Prompt Formatting. Our evaluation spans two complementary domains:

1. *Coordinate-Based Spatial Grounding Tasks (9 Benchmarks)*: Tasks that require the model to output 2D pixel coordinates or ordered coordinate sequences in image space. This suite covers **affordance and interaction pointing** on Part-Affordance-2K [60], RoboAfford [22], VABench-Point [59], and Where2Place [58]; **referring and spatial localization** on PIOBench [54], RefSpatial-Bench [62], and RoboRefit [34]; and **trajectory prediction** on ShareRobot-Trajectory [24] and VABench-Visual-Trace [59].
2. *Text-Based Embodied Understanding & Reasoning Tasks (11 Benchmarks)*: Tasks where the model responds in natural language, such as multiple-choice answers, numerical estimates, or textual action plans. These benchmarks encompass **scene spatial perception and relational QA** on BLINK [19], CV-Bench [47], EmbSpatial [16], and SAT [42]; **multi-view and video**

Task Modality	Training Data Sources
General Multimodal Cognition	
General Understanding	Visual7W [64], Visual Genome [27]
Spatial Understanding	RefSpatial [62], RoboSpatial [45], RoboPoint [58]
General Localization	EgoObjects [63], LVIS [21], PixMo-Points [13], V3Det [49]
Embodied Physical Grounding & Planning	
Embodied Understanding	FSD [59], Robo2VLM [9]
Grounding & Affordance	RAGNet [51], AffGrasp [11], HOI4D [33], RoboAfford [22], ShareRobot [24], GraspNet [17], VL-Grasp [34], GraspClutter6D [5]
Trajectory Prediction	MolmoAct [28], A_0 [53]
Reasoning & Planning	RoboVQA [43], ShareRobot [24]
Total Multi-Task Mixture: ∼12.0 Million Training Samples	

Table 1. **Comprehensive multi-task training mixture (∼12M samples) for PhysMoE**. Datasets are structured across general multimodal cognition and embodied physical grounding. For composite suites (*e.g.*, ShareRobot [24]), we incorporate their respective domain subsets. Point and bounding-box annotations in affordance and contact tasks are converted into binary masks via SAM3 [26, 41] for continuous Mask-SDF supervision.

spatio-temporal reasoning on MindCube [50], ViewSpatial [29], and VSI-Bench [56]; and **physical dynamics and embodied task planning** on COSMOS [4], ERQA [1], ERQA-PLUS [55], and RoboVQA [43].

Because PhysMoE directly outputs complete, continuous 2D coordinate units rather than character-by-character token strings, it does not require verbose prompt constraints specifying coordinate output formatting. Accordingly, we omit verbose format instructions on coordinate-based tasks and simply append `Type: point/bbox/trajectory` to specify the target output for our model.

Metrics. All scores are reported on a standard 0–100 scale (higher is better). Point grounding tasks are evaluated using micro-averaged $\mathrm{F}_1 = 2\mathrm{P}_\mu\mathrm{R}_\mu/(\mathrm{P}_\mu+\mathrm{R}_\mu)\times 100$, where precision $\mathrm{P}_\mu = \sum_i H_i / \sum_i P_i$ measures the proportion of predicted points falling inside valid ground-truth masks, and recall $\mathrm{R}_\mu = \sum_i \min(M_i, G_i)/\sum_i G_i$ measures the proportion of covered targets (P_i, G_i are prediction and target counts; H_i, M_i are hit and matched counts for sample i) For trajectory benchmarks (ShareRobot-Trajectory and VABench-Visual-Trace), following standard evaluation protocols, trajectory accuracy is computed as $100 \times (1 - \text{mean_RMSE}/1000)$. The best and second-best scores among parameter-comparable models are highlighted in dark blue ( ) and light blue ( ), respectively.

Baselines. We evaluate PhysMoE at both 4B and 9B scales against representative open-source and published embodied foundation models. At the 4B scale, we compare with the base backbone Qwen3.5 (4B), RynnBrain 1.0 (4B) [30], and RoboBrain 2.5 (4B) [46]. At the 8B–9B (and MoE) scale, our competitors include RoboBrain 2.5 (8B) [46], RynnBrain 1.1 (9B) [30], Cosmos3-Nano (8B) [3], Embodied-R1.5 (8B) [61], and the large-scale mixture-of-experts model HY-Embodied-VLM-1.0 (30A3B) [23]. Note that some results for Embodied-R1.5, Cosmos3-Nano, and HY-Embodied-VLM in Tabs. 2 and 3 are adopted from *HY-Embodied-VLM-1.0* [23], while all other models are evaluated locally under unified benchmark protocols.

4.2. Main Quantitative Results and Analysis

Tab. 2 and Tab. 3 evaluate PhysMoE across 9 coordinate-based spatial grounding tasks and 11 text-based embodied understanding and reasoning tasks, validating two key properties of our framework:

1. Strong Gains on Coordinate-Based Spatial Grounding. As shown in Tab. 2, PhysMoE substantially improves over its base models at both scales—rising from 56.3 to **67.8** on 4B (over Qwen3.5-4B) and from 64.1 to **74.2** on 9B (+15.7% relative over RynnBrain 1.1)—while also outperforming specialized embodied VLMs including the 3× larger 30B HY-Embodied-VLM (69.2). The advantage of our coordinate encoding is most evident on trajectory prediction, the most pixel-context-rich task: each waypoint is predicted conditioned on previously generated ones, so the model must read its own coordinates back as context. Since each coordinate is a single token embedded in the visual encoder's positional space, earlier waypoints stay aligned with image patches instead of being fragmented into digit tokens. Accordingly, PhysMoE-9B ranks first on both ShareRobot-Traj (**87.1**) and VABench-Trace (**90.2**).

2. PhysMoE Enhances Spatial Grounding While Preserving Text-Based Reasoning. Comparing Tab. 2 and Tab. 3 relative to each model's initialization backbone demonstrates that our PhysMoE architecture prevents coordinate regression from interfering with pre-trained linguistic representations. By sharing lower layers for multimodal context while isolating spatial regression gradients within $\mathrm{FFN}_{\mathrm{pix}}$ across the top K layers, PhysMoE achieves substantial gains in coordinate grounding (+11.5 points on 4B and +10.1 points on 9B) while maintaining text-based reasoning performance on par with its base models: PhysMoE-4B slightly improves over Qwen3.5-4B (62.7 *vs.* 61.3), and PhysMoE-9B closely matches RynnBrain 1.1 (70.9 *vs.* 71.4).

4.3. Ablation Studies

To validate our core design choices, we conduct ablation analyses on the 4B model variant (PhysMoE-4B), investigating continuous coordinate token encoding and mask-

Benchmark	Qwen3.5 (4B)	Rynn1.0 (4B)	RoboBrain (4B)	Ours (4B)	RoboBrain (8B)	Rynn1.1 (9B)	Cosmos3 (8B)	Embodied (8B)	HY (30A3B)	Ours (9B)
Part-Affordance	17.9	15.5	34.8	57.2	25.1	42.9	32.2	82.6	63.7	82.2
PIOBench	49.4	58.5	56.8	52.1	60.2	62.2	54.4	61.6	65.3	60.7
RefSpatial-Bench	39.7	56.8	53.3	53.8	54.9	62.0	44.4	52.4	53.4	55.2
RoboAfford	65.6	73.1	77.4	73.5	72.5	72.8	76.2	60.6	71.5	74.8
RoboRefit	81.0	61.0	83.5	82.0	82.4	81.9	55.4	77.2	88.2	82.0
VABench-Point	37.8	2.3	45.8	46.7	9.6	13.2	45.2	61.4	59.7	62.7
Where2Place	58.7	70.9	39.3	71.9	69.0	72.7	71.0	73.0	65.0	72.6
ShareRobot-Traj	77.5	81.7	86.8	85.9	85.2	80.9	65.5	69.2	76.7	87.1
VABench-Trace	79.5	84.3	83.5	87.2	86.7	87.9	81.6	89.8	79.7	90.2
Average	56.3	56.0	62.4	67.8	60.6	64.1	58.4	69.8	69.2	74.2

Table 2. **Comprehensive Evaluation on Coordinate-Based Spatial Grounding Tasks**. Scores are on a 0–100 scale (↑). and denote the best and second-best results among parameter-comparable models, respectively. Model abbreviations: Rynn1.0 (RynnBrain1.0), Rynn1.1 (RynnBrain1.1), RoboBrain (RoboBrain2.5), Cosmos3 (Cosmos3-Nano), Embodied (Embodied-R1.5), HY (HY-Embodied-VLM-1.0).

Benchmark	Qwen3.5 (4B)	Rynn1.0 (4B)	RoboBrain (4B)	Ours (4B)	RoboBrain (8B)	Rynn1.1 (9B)	Cosmos3 (8B)	Embodied (8B)	HY (30A3B)	Ours (9B)
BLINK	78.8	82.2	78.0	75.5	80.9	85.0	82.4	77.8	87.3	80.6
CV-Bench	86.2	89.2	87.4	78.2	87.6	87.7	88.0	86.8	89.7	88.7
EmbSpatial	77.3	80.6	75.9	68.9	77.9	81.9	80.0	76.0	82.7	81.9
MindCube	35.0	55.5	34.0	46.2	29.3	81.1	32.8	27.9	70.0	79.0
SAT	69.3	65.3	64.0	71.3	66.0	76.7	54.0	60.7	78.0	78.0
VSI-Bench	53.0	65.7	41.5	56.1	44.1	66.0	50.4	59.2	58.9	65.3
ViewSpatial	39.7	54.5	41.6	46.2	39.1	55.2	52.0	43.7	53.3	52.1
COSMOS	60.8	58.2	53.9	60.8	58.4	64.5	67.1	68.6	66.9	62.9
ERQA	46.0	43.8	40.2	42.8	44.8	43.0	45.0	37.3	60.8	47.0
ERQA-PLUS	82.8	77.3	81.8	80.6	81.0	83.6	83.6	80.7	82.2	80.8
RoboVQA	45.4	61.9	38.9	62.9	43.7	60.9	50.2	60.8	41.2	63.6
Average	61.3	66.7	57.9	62.7	59.3	71.4	62.3	61.8	70.1	70.9

Table 3. **Evaluation on Text-Based Embodied Understanding & Reasoning Tasks**. and denote the best and second-best results among parameter-comparable models, respectively.

based supervision objectives.

1. Analyzing Coordinate Token Encoding via Learnable Gating. We initially considered three candidate schemes for encoding a coordinate token $(x, y) \in [0, 1]^2$: reusing the visual encoder's learned positional embeddings ($\mathbf{e}_{\text{pe}}$), sinusoidal Fourier frequency encoding ($\mathbf{e}_{\text{f}}$), and local visual patch features sampled at that coordinate ($\mathbf{e}_{\text{i}}$). To allow the model to automatically learn which encoding scheme is most suitable, we combine the three candidates via a learnable gated fusion layer during training:

$$\begin{aligned} \tilde{\mathbf{e}}_{\text{fused}} &= w_{\text{pe}}\mathbf{e}_{\text{pe}} + w_{\text{f}}\mathbf{e}_{\text{f}} + w_{\text{i}}\mathbf{e}_{\text{i}}, \\ \mathbf{e}_{\text{fused}} &= \frac{\|\mathbf{e}_{\text{pe}}\|_2}{\|\tilde{\mathbf{e}}_{\text{fused}}\|_2}\tilde{\mathbf{e}}_{\text{fused}}, \end{aligned} \tag{10}$$

where $\mathbf{e}_{\text{pe}}$ projects the interpolated visual positional embedding $\mathcal{P}_{\text{vis}}$ (Eq. (5)); $\mathbf{e}_{\text{f}}$ maps 12-frequency sinusoidal Fourier features of (x, y) (frequencies $2^k\pi$ for $k = 0, \ldots, 11$) through a two-layer MLP; and $\mathbf{e}_{\text{i}}$ applies an affine-free LayerNorm and linear projection to pre-merger visual patch features sampled at (x, y). Crucially, we rescale $\tilde{\mathbf{e}}_{\text{fused}}$ to match the ℓ_2 norm of $\mathbf{e}_{\text{pe}}$ so that the fusion branches only adjust the embedding direction without altering the magnitude scale expected by the language model, making the gating weights directly reflect relative contributions.

Analyzing the learned gating weights and branch norm shares ($\|w_k\mathbf{e}_k\|_2 / \sum_j \|w_j\mathbf{e}_j\|_2$) at convergence reveals distinct behaviors:

Supervision Objective	Part-Afford.	RoboRefit	ShareRobot-Traj	RoboAfford
(a) Without Mask (Centroid ℓ_1)	55.5	79.2	85.5	72.3
(b) With Mask-SDF (Ours)	**57.2**	**82.0**	**85.9**	**73.5**

Table 4. **Ablation Study on Supervision Objectives (Evaluated on 4B Model)**. Comparison of single-point centroid regression versus our differentiable Mask-SDF supervision across affordance grounding, referring expressions, and trajectory prediction.

- **Learned Visual PE Dominates** ($> 97\%$): Reusing the visual backbone's learned positional embeddings accounts for over 97% of the representation magnitude, as $\mathcal{P}_{\text{vis}}$ inherently encodes 2D metric geometry aligned with visual tokens.
- **Image Features Provide Marginal Context** ($\sim 3\%$): Pre-merger visual patch features contribute only $\sim 3\%$; while offering semantic appearance cues, patch-level features lack the fine spatial resolution required for exact coordinate localization.
- **Fourier Encoding Fails to Contribute** ($< 1\%$): The learned Fourier weight vanishes to less than 1% ($w_{\text{f}} \approx 0$), as generic sinusoidal encodings operate in an unaligned domain disjoint from the visual representation manifold.

These results confirm that directly reusing the visual backbone's learned PE provides the most effective coordinate representation, rendering auxiliary Fourier embeddings and patch feature injections unnecessary.

2. Effectiveness of Differentiable ESDF Supervision on Pointing. Under identical training data and architecture, Tab. 4 compares single-point centroid regression (ℓ_1) against our differentiable ESDF supervision. By accommodating arbitrary interior contact points on multi-contact affordances and irregular geometries rather than forcing predictions toward a single centroid, ESDF supervision consistently improves RoboRefit ($79.2\% \rightarrow$ **82.0%**), Part-Affordance ($55.5\% \rightarrow$ **57.2%**), and RoboAfford ($72.3\% \rightarrow$ **73.5%**). Furthermore, although ESDF is not directly applied to trajectory sequences, ShareRobot-Traj also improves ($85.5\% \rightarrow$ **85.9%**) due to more accurate localization of trajectory start and target points.

4.4. Inference Efficiency and Pixel Throughput

In physical robot manipulation, low latency and high inference throughput are essential for responsive, closed-loop execution. In Tab. 5, we benchmark the pixel inference speed across representative embodied foundation models on an NVIDIA RTX 5090 GPU under identical evaluation settings. To construct a representative evaluation suite, we sample 1,000 QA instances from the benchmarks where the primary target output consists of 2D pixel coordinates (spanning point grounding, bounding boxes, and motion trajectories).

Here, the evaluation metric is defined as **pixel inference speed** measured in **samples per second (samples/s)**, where a complete coordinate tuple alongside its formatting delimiters constitutes one sample. As shown in Tab. 5, existing autoregressive models serialize coordinates into multiple discrete tokens, requiring iterative autoregressive decoding passes that bottleneck throughput to only 1.58–5.28 samples/s.

Model	Coord. Decoding	Throughput (samples/s) ↑
RynnBrain1.0-4B [30]	Autoregressive Tokens	5.28
RoboBrain2.5-4B [46]	Autoregressive Tokens	3.41
RynnBrain1.1-9B [30]	Autoregressive Tokens	2.61
EmbodiedR1.5-8B [61]	Autoregressive Tokens	2.81
RoboBrain2.5-8B [46]	Autoregressive Tokens	1.58
Ours-4B	Direct Regression	**31.88** ($6.04\times$)
Ours-9B	Direct Regression	**23.34** ($8.31\times$)

Table 5. **Pixel Inference Speed Comparison**. Measured in samples/second (samples/s ↑). Speedups in parentheses are computed conservatively relative to the fastest competing baseline at each model scale.

In contrast, PhysMoE represents each 2D coordinate as a single token and predicts it in one decoding step via the coordinate head, bypassing multi-token character serialization entirely. Notably, even when benchmarked against the fastest competing baseline in each parameter class, **Ours-9B** attains a pixel inference throughput of **23.34 samples/s** ($8.31\times$ over EmbodiedR1.5-8B), while **Ours-4B** reaches **31.88 samples/s** ($6.04\times$ over RynnBrain1.0-4B). This substantial acceleration provides critical responsiveness for high-frequency robotic control loops without sacrificing spatial precision.

5. Conclusion

In this paper, we introduced PhysMoE, an embodied vision-language model that harmonizes continuous physical manipulation with high-level multimodal reasoning. By decoupling top-layer FFNs via a Semi-MoE framework while preserving shared self-attention, PhysMoE eliminates gradient interference between categorical text generation and geometric regression. Furthermore, by reusing pre-trained visual positional embeddings for continuous coordinate encoding and formulating differentiable Mask-SDF supervision for multi-peak affordances, our model achieves sub-pixel precision without auxiliary mask decoders. Extensive evaluations across 20 benchmarks validate that PhysMoE achieves state-of-the-art trajectory accuracy and outstanding affordance grounding while preserving conversational reasoning fidelity.

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